

Module and Assessment Guide

Module Title:	Recommender Systems	
Module Code:	B9AI003	
Programme:	MSc in AI Postgraduate Diploma in Science in Artificial Intelligence	
Academic Year:	2026	
Delivery Type F/PT	Full time	
Delivery Mode	Classroom	
Lecturer Name(s):	Nitya Govindaraju	Email: nitya.govindaraju@dbs.ie

Learning Outcomes:

On completion of this module, learners will be able to:

1.	Evaluate the principles and methodologies of diverse recommender systems, such as collaborative, content-based, demographic, utility-based, knowledge-based, and hybrid systems, to inform advanced design and implementation.
2.	Apply evaluation and experimentation techniques specific to recommender systems, utilizing metrics such as NDCG, MAP, A/B testing, and longitudinal testing for sustained engagement scenarios.
3.	Critically assess ethical and social implications of recommendation systems, including bias detection and mitigation, privacy-preserving methods, and case studies on fairness and bias issues in recommendations.
4.	Critically evaluate Recommender Systems within the context of other Machine Learning approaches.
5.	Design, develop, and evaluate a recommender system for a practical use case, addressing key challenges and applying Responsible AI principles in the design and deployment of AI systems.
6.	Identify the robustness aspects of Recommender Systems, such as shilling systems, attack models, and their defences, learning to rank, multi-armed bandits, group systems, multi-criteria systems, and active learning systems will be learned.

Learning Activities:

Each week will consist of a number of different activities:

- Presentation
- Practical Demonstration / Hands-on Lab
- Group Discussion
- Individual Practice and Coding Tasks
- Group Work / Team Activities
- Oral Presentation / Peer Feedback

Week by Week Teaching Plan:

Week	LO	Topic	Mode	Activities
1	1	Introduction to Recommender Systems: architectures, methods, user profiling, personalization	Blended	Presentation, discussion, demo, hands-on practice
2	1,5	Collaborative Filtering (1): memory-based approaches, similarity	Blended	Presentation, demo, coding lab, practice
3	1,5	Collaborative Filtering (2): model-based approaches (matrix factorization concept)	Blended	Demo, implementation, group discussion
4	1,5	Content-Based Filtering: feature engineering, TF-IDF style intuition	Blended	Practical demo, hands-on, group activity
5	1,5	Knowledge-based & Utility-based Recommenders	Blended	Presentation, case studies, design exercise
6	1,2,5	Hybrid Recommenders + Cold Start + Scalability	Blended	Group discussion, coding lab, applied strategies
7	2,5	Evaluation & Experimentation: offline metrics (NDCG, MAP), A/B testing overview	Blended	Lab + assessment support
8	6	Generative AI in Recommendations: GANs/VAEs ideas, comparison with traditional approaches	Blended	Practical lab: simple generative-based concept
9	3	Ethical & Social Implications: bias, fairness, echo chambers, privacy methods	Blended	Case analysis + discussion
10	4	Explainable AI (XAI): SHAP/LIME + interpretable outputs	Blended	Lab + What-If Tool demo
11	1,5	Real-world applications: e-commerce/media/social case studies + system design	Blended	Industry problem analysis + group progress

Readings:

Core Texts (Primary Reading)

- Ping, D. (2024) *The Machine Learning Solutions Architect Handbook: Practical strategies and best practices on the ML lifecycle, system design, MLOps, and generative AI*. Birmingham: Packt Publishing.

Supplementary Reading

- ACM Recommender Systems conference papers (privacy, ethics, fairness, XAI).
- Google What-If Tool documentation.
- SHAP documentation and guides.
- LIME documentation and guides.
- Molnar, C. (2022) *Interpretable Machine Learning*. 2nd ed. Leanpub.
- Schrage, M. (2020) *Recommendation Engines*. MIT Press.
- Moolayil, J. (2019) *Learn Keras for Deep Neural Networks*. Apress.

Assessment:

Method of Assessment	Percentage Weighting	Learning Outcome Being Assessed	Date of Submission
CA 1: Group Project (Model Development & Evaluation)	60%	1,2,5,6	Week 7
CA 2: Individual Report (Explainability + Ethics)	40%	3,4,6	Week 14

*See Moodle Assessment link for assessment due dates

Overview of Assessment:

CA 1: Group Project on Model Development and Evaluation Techniques (60%)

Assessment Title & Description:	Group Project
Task:	Students work in groups to design, develop, and evaluate a recommender system for a chosen practical use case such as: • e-commerce product recommendation • media/movie recommendation • music recommendation • social networking content recommendation The project must include: 1. Recommender approach selection and justification 2. Model development (baseline + improved version) 3. Evaluation using appropriate metrics 4. Addressing challenges such as cold start, scalability, robustness 5. Inclusion of a section on generative AI integration and comparison with traditional approaches 6. Presentation + report submission Submission Requirements • Group report • Python notebooks (or equivalent code submission) • Group presentation
MIMLOs being assessed:	1,2,5,6
Individual/Group:	Group
Assessment Weighting:	60%
Issue Date:	Week 2

Submission Date: (All assignments must be submitted through Moodle)	Week 7
Feedback Date:	Week 9

CA 2: Individual Report on Explainability and Ethics in Recommender Systems

(40%)

Assessment Title & Description:	Individual Report
Task:	Each learner will evaluate a recommender system using XAI methods. Report must include: • Selected dataset/system • Explainability method used (SHAP / LIME) • Explanation outputs and interpretation • Ethical analysis: bias/fairness considerations and reflection
MIMLOs being assessed:	3,4,6
Individual/Group:	Individual
Assessment Weighting:	40%
Issue Date:	Week 7
Submission Date: (All assignments must be submitted through Moodle)	Week 14
Feedback Date:	Week 15

Recommended Software / Tools

- Python
- TensorFlow / PyTorch
- Scikit-learn
- Surprise
- Pandas, NumPy
- Matplotlib, Seaborn
- SHAP, LIME
- Jupyter Notebook
- GitHub (group collaboration)

Notes on Generative AI Use (Assessment Guidance)

- brainstorming and structuring ideas
- improving outlines and planning
- AI content is not allowed in the final submission (students must write in their own words).